IGCSE · Cambridge (CIE) · Maths

2 hours 14 questions

Calculator Questions

Cumulative Frequency

Cumulative Frequency / Drawing Cumulative Frequency Diagrams / Interpreting Cumulative Frequency Diagrams

Easy (3 questions)	/15
Medium (6 questions)	/41
Hard (5 questions)	/43
Total Marks	**/99**

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Easy Questions

1 (a) Sue works for a company that delivers parcels.

One day the company delivered 80 parcels.
The table shows information about the weights, in kg, of these parcels.

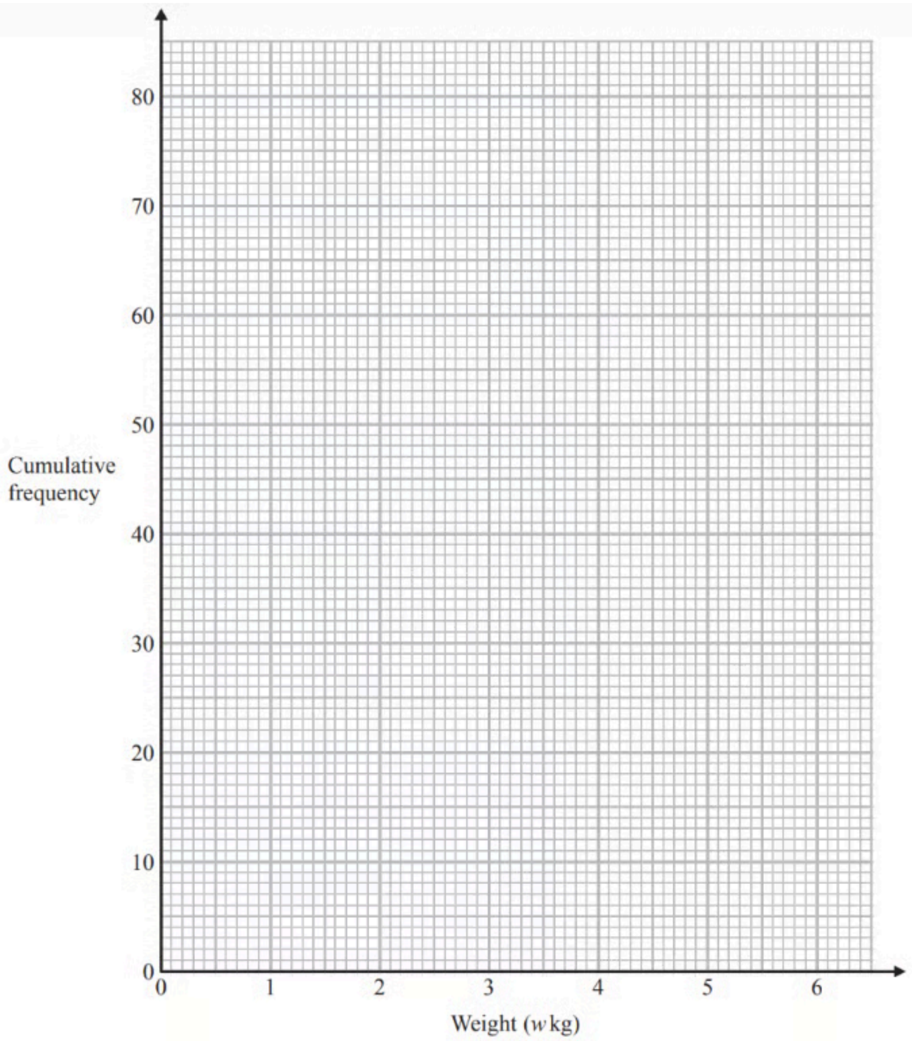
Weight (w kg)	**Frequency**
$0 < w \leqslant 1$	19
$1 < w \leqslant 2$	17
$2 < w \leqslant 3$	15
$3 < w \leqslant 4$	12
$4 < w \leqslant 5$	10
$5 < w \leqslant 6$	7

Complete the cumulative frequency table.

Weight (w kg)	Cumulative frequency
$0 < w \leqslant 1$	
$0 < w \leqslant 2$	
$0 < w \leqslant 3$	
$0 < w \leqslant 4$	
$0 < w \leqslant 5$	
$0 < w \leqslant 6$	

(1 mark)

(b) On the grid opposite, draw a cumulative frequency graph for your table.



(2 marks)

(c) Sue says,

"75 % of the parcels weigh less than 3.4 kg."

Is Sue correct?
You must show how you get your answer.

(3 marks)

2 (a) The grouped frequency table gives information about the times, in minutes, that 80 office workers take to get to work.

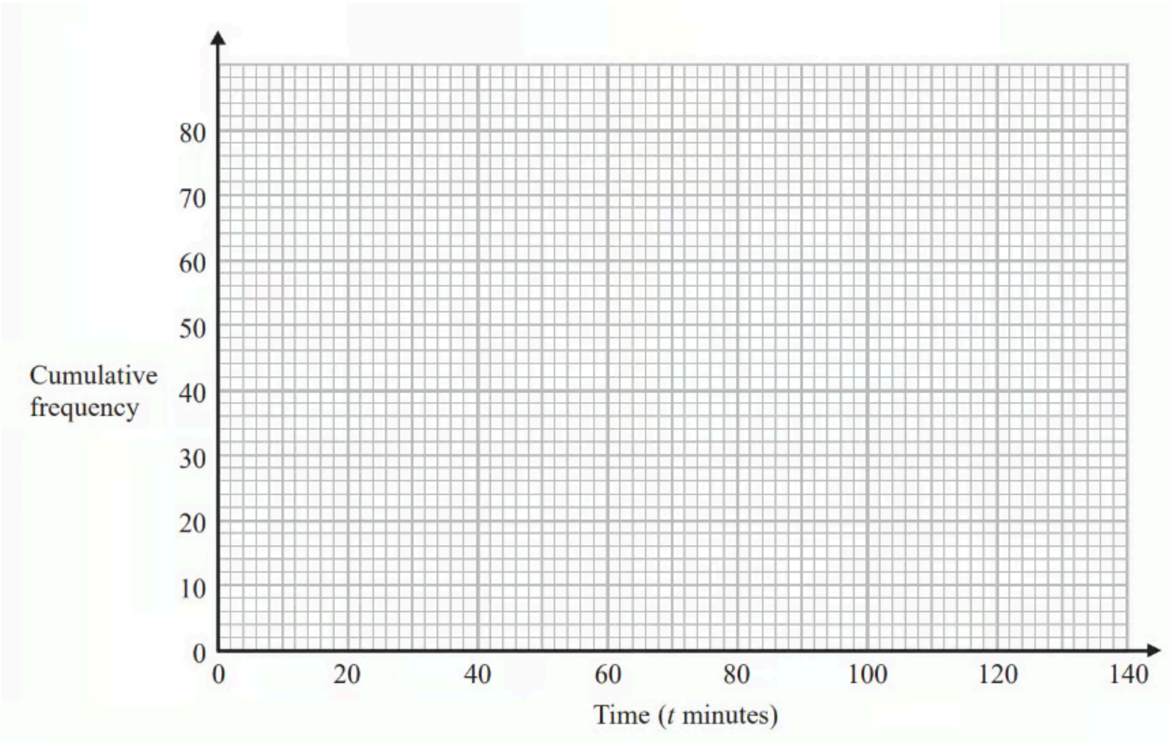
Time (t minutes)	**Frequency**
$0 < t \leqslant 20$	5
$20 < t \leqslant 40$	30
$40 < t \leqslant 60$	20
$60 < t \leqslant 80$	15
$80 < t \leqslant 100$	8
$100 < t \leqslant 120$	2

Complete the cumulative frequency table.

Time (t minutes)	**Cumulative frequency**
$0 < t \leqslant 20$	
$0 < t \leqslant 40$	
$0 < t \leqslant 60$	
$0 < t \leqslant 80$	
$0 < t \leqslant 100$	
$0 < t \leqslant 120$	

(1 mark)

(b) On the grid, draw the cumulative frequency graph for this information.

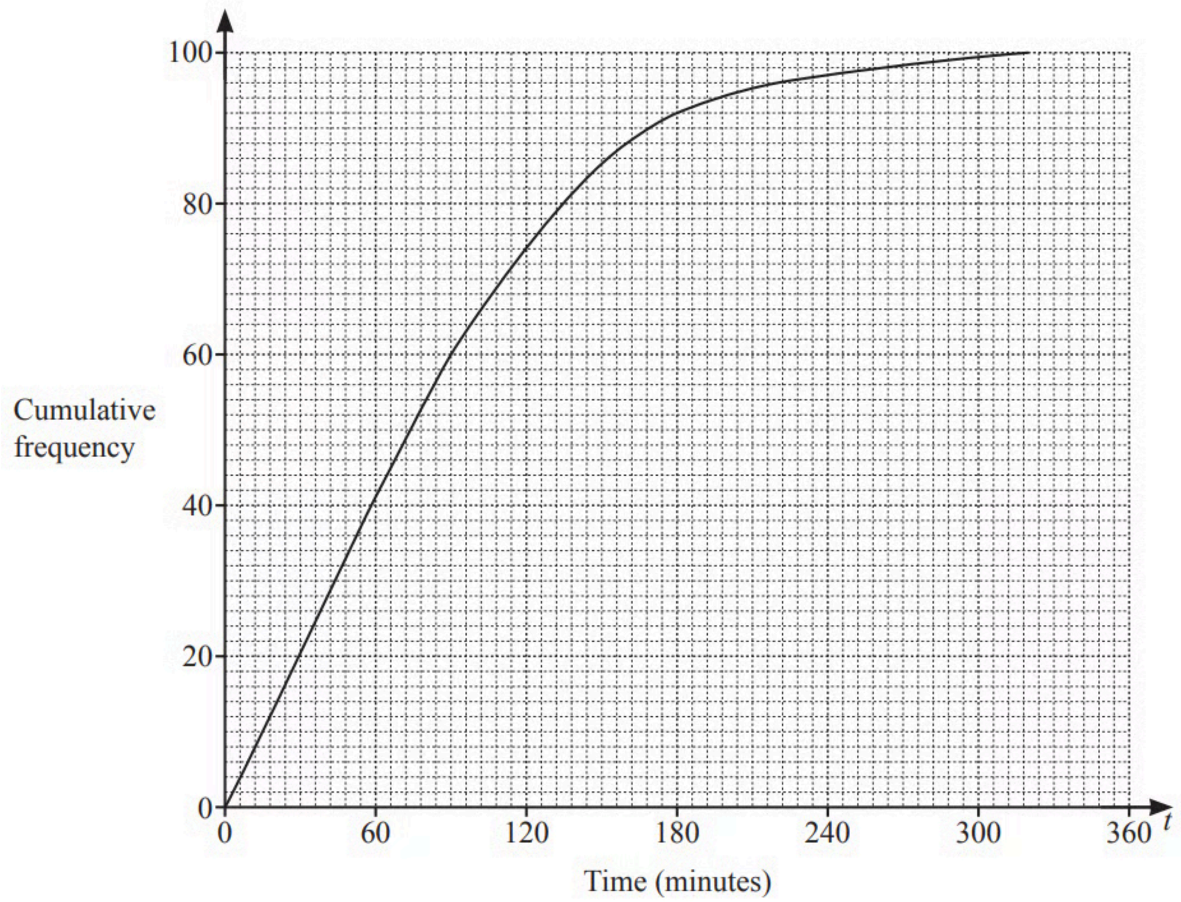


(2 marks)

(c) Use your graph to find an estimate for the percentage of these office workers who take more than 90 minutes to get to work.

(3 marks)

3 The cumulative frequency graph shows the times, in minutes, each of 100 children spent exercising in one week.



Use the cumulative frequency diagram to find an estimate of

i) the 60th percentile,

.... min [1]

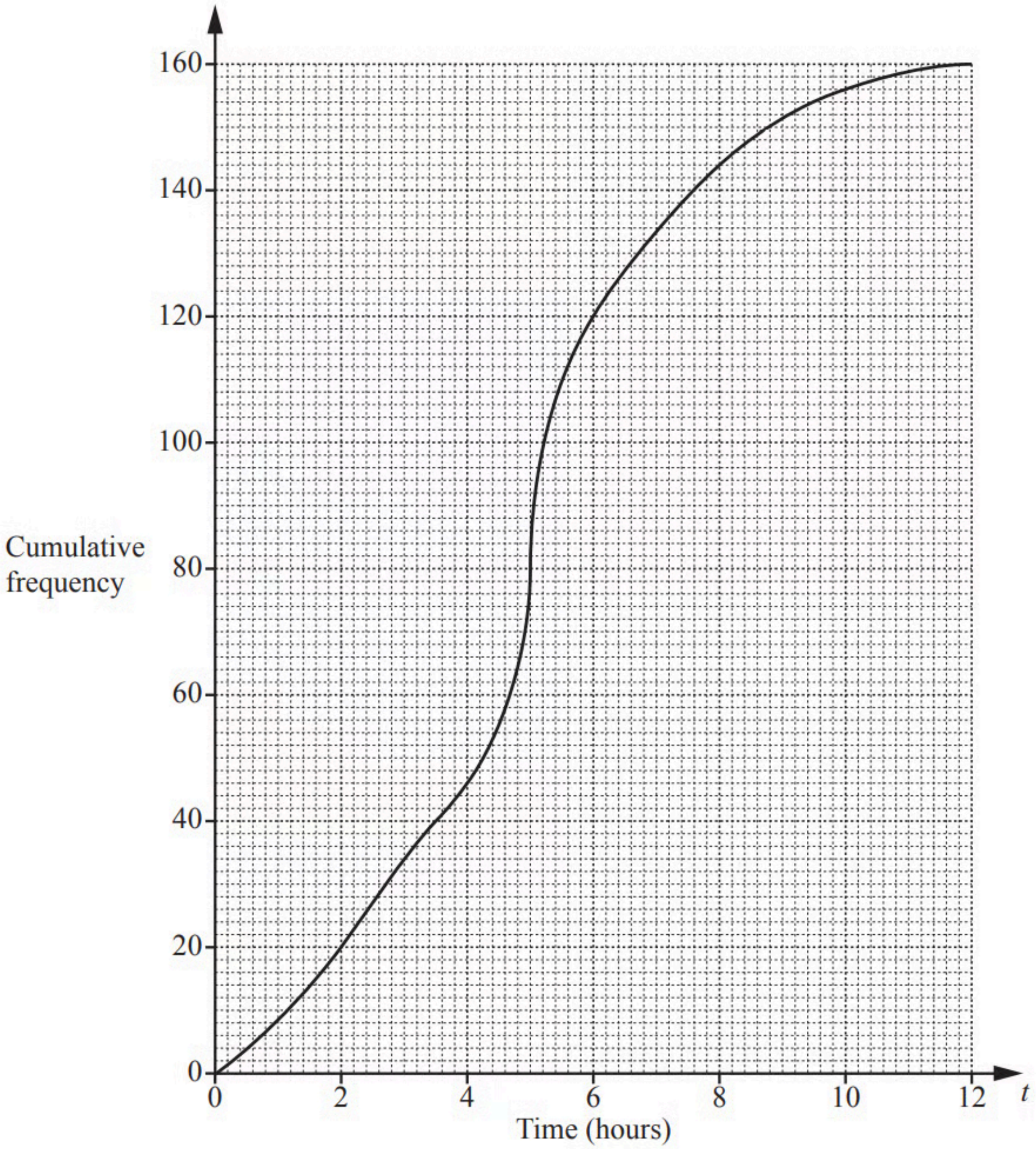
ii) the number of children who spent more than 3 hours exercising.

[2]

(3 marks)

Medium Questions

1 (a) 160 students record the amount of time, t hours, they each spend playing computer games in a week.
This information is shown in the cumulative frequency diagram.



Use the diagram to find an estimate of

i) the median,

.... hours [1]

ii) the interquartile range.

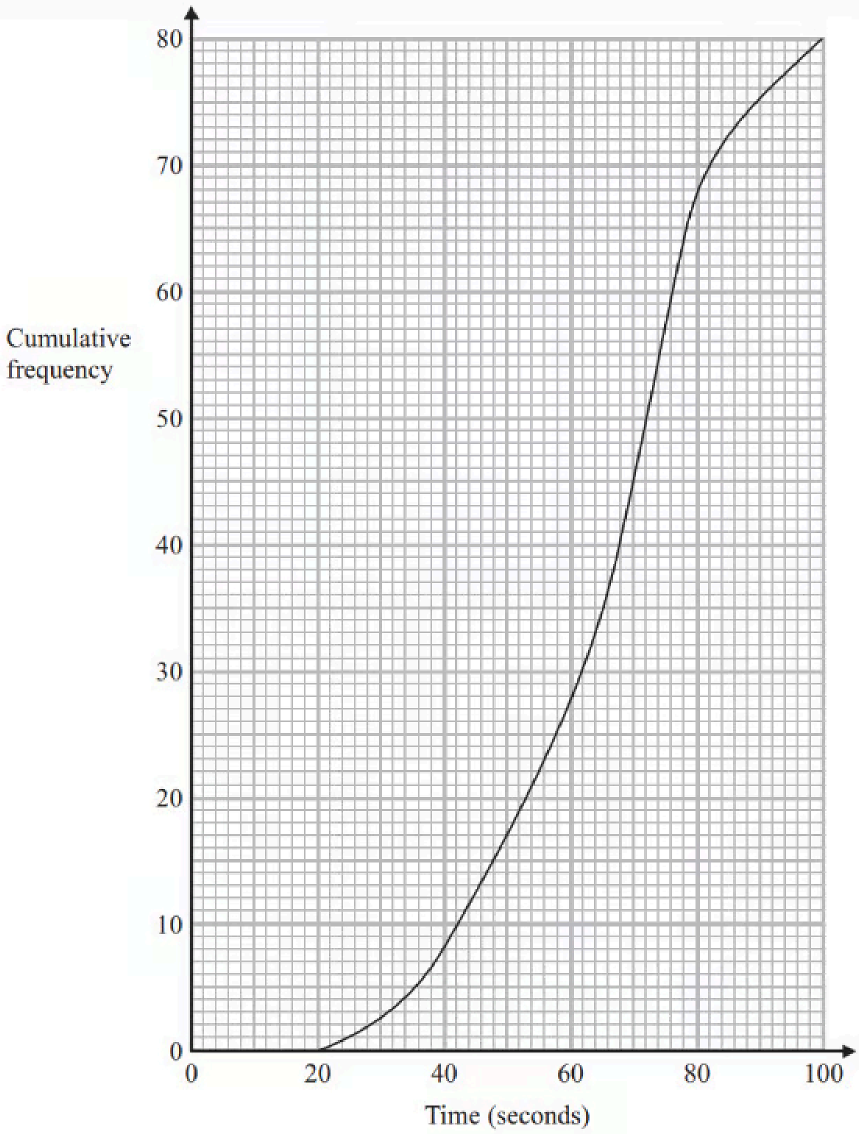
.... hours [2]
(3 marks)

(b) Use the diagram to complete this frequency table.

Time (t hours)	$0 < t \leq 2$	$2 < t \leq 4$	$4 < t \leq 6$	$6 < t \leq 8$	$8 < t \leq 10$	$10 < t \leq 12$
Frequency	20			24	12	4

(2 marks)

2 (a) The cumulative frequency graph shows information about the times 80 swimmers take to swim 50 metres.



Use the graph to find an estimate for the median time.

(1 mark)

(b) A swimmer has to swim 50 metres in 60 seconds or less to qualify for the swimming team.

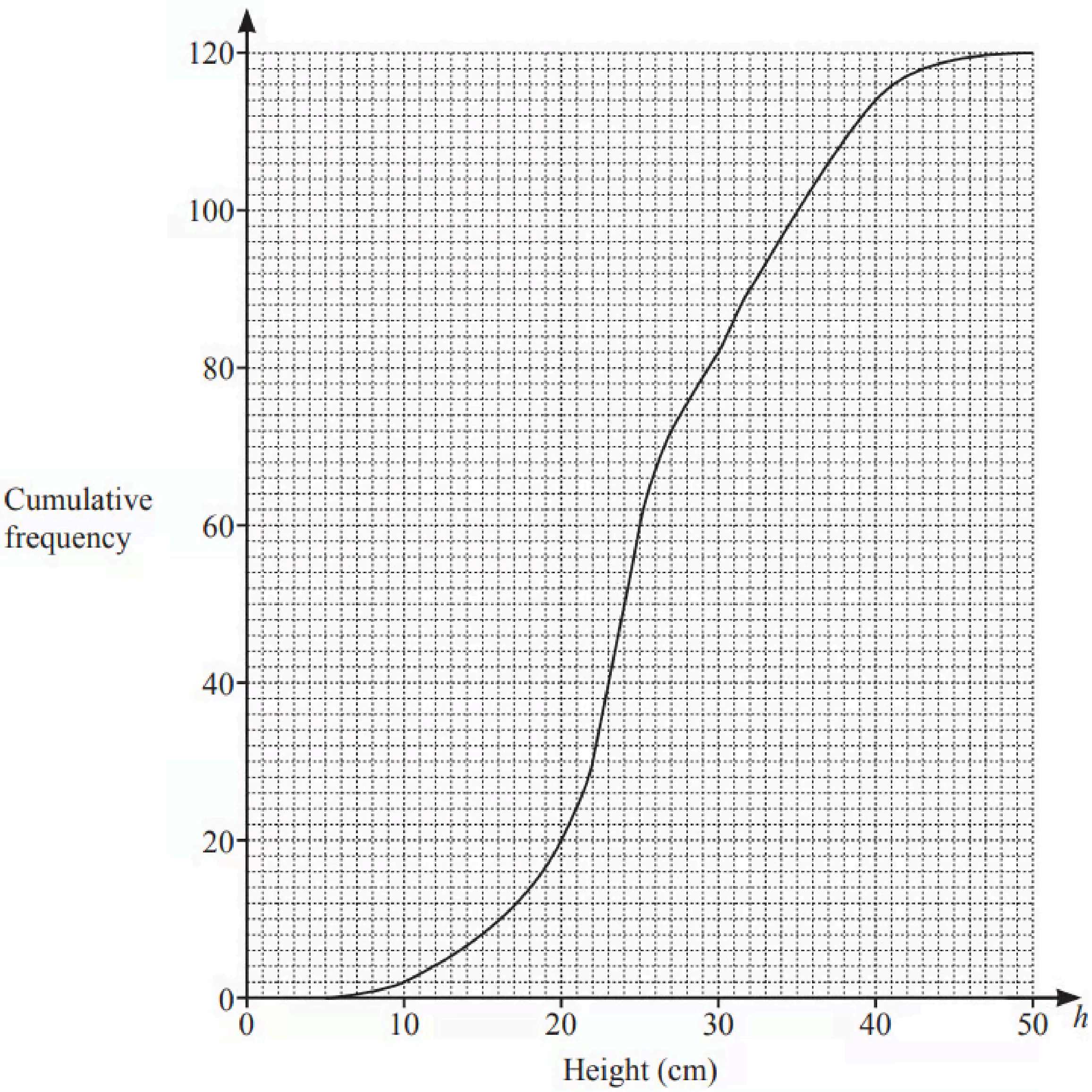
The team captain says,

"More than 25% of swimmers have qualified for the swimming team."

Is the team captain right?
You must show how you got your answer.

(3 marks)

3 The height, h cm, of each of 120 plants is measured.
The cumulative frequency diagram shows this information.



Use the cumulative frequency diagram to find an estimate of

i) the median,

.... cm [1]

ii) the interquartile range,

.... cm [2]

iii) the 60th percentile,

.... cm [1]

iv) the number of plants with a height greater than 40cm.

[2]

(6 marks)

4 (a) A school nurse records the height, h cm, of each of 180 children.
The table shows the information.

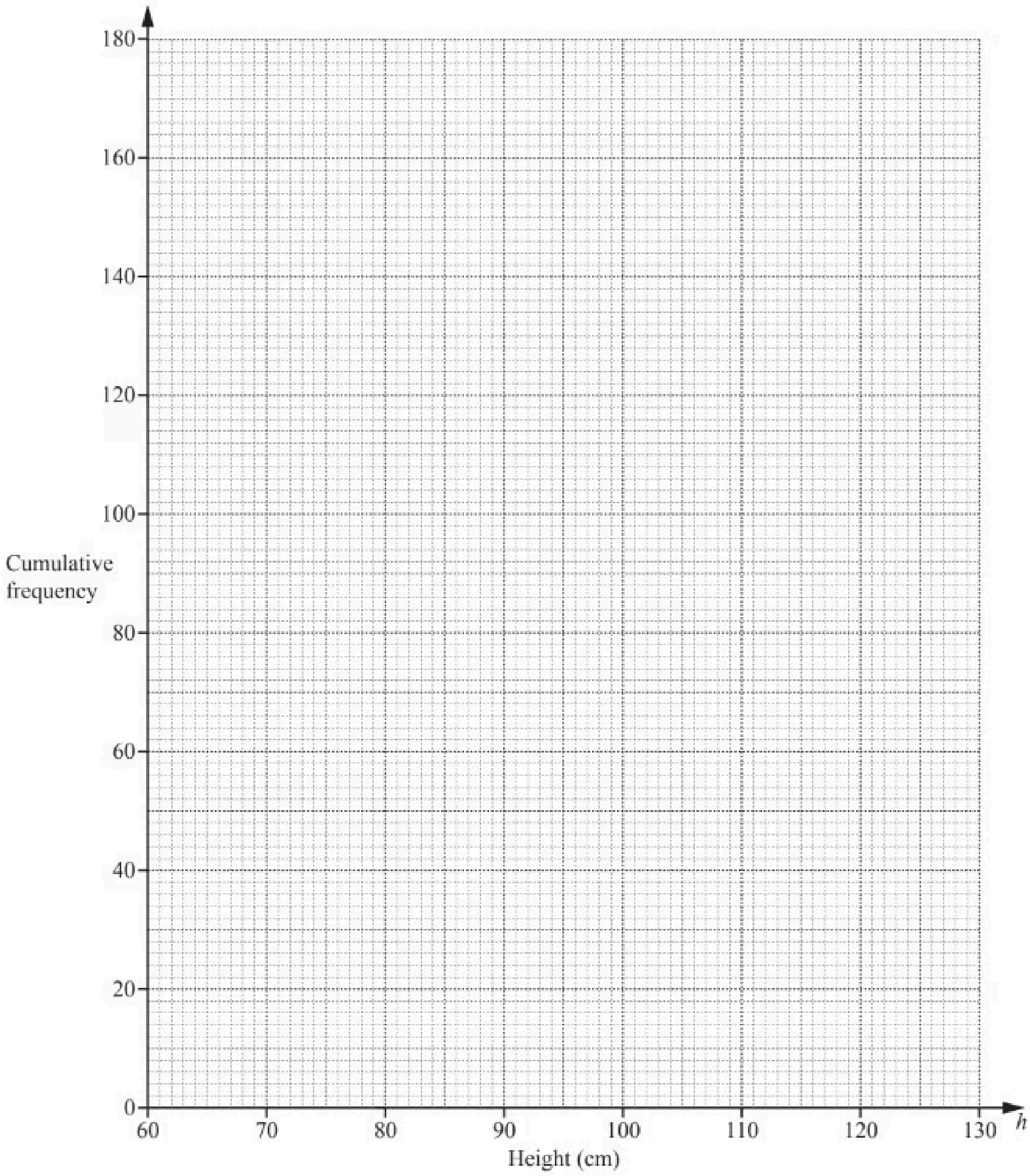
Height (h cm)	$60 < h \le 70$	$70 < h \le 90$	$90 < h \le 100$	$100 < h \le 110$	$110 < h \le 115$	$115 < h \le 125$
Frequency	8	26	35	67	28	16

Complete the cumulative frequency table below.

Height (h cm)	$h \le 70$	$h \le 90$	$h \le 100$	$h \le 110$	$h \le 115$	$h \le 125$
Cumulative Frequency						180

(2 marks)

(b) On the grid below, draw a cumulative frequency diagram.



(3 marks)

(c) Use your cumulative frequency diagram to find an estimate of

i) the interquartile range,

.... cm [2]

ii) the 7 0^{th} percentile,

.... cm [2]

iii) the number of children with height greater than 106 cm.

[2]

(6 marks)

5 The time taken for each of 120 students to complete a cooking challenge is shown in the table.

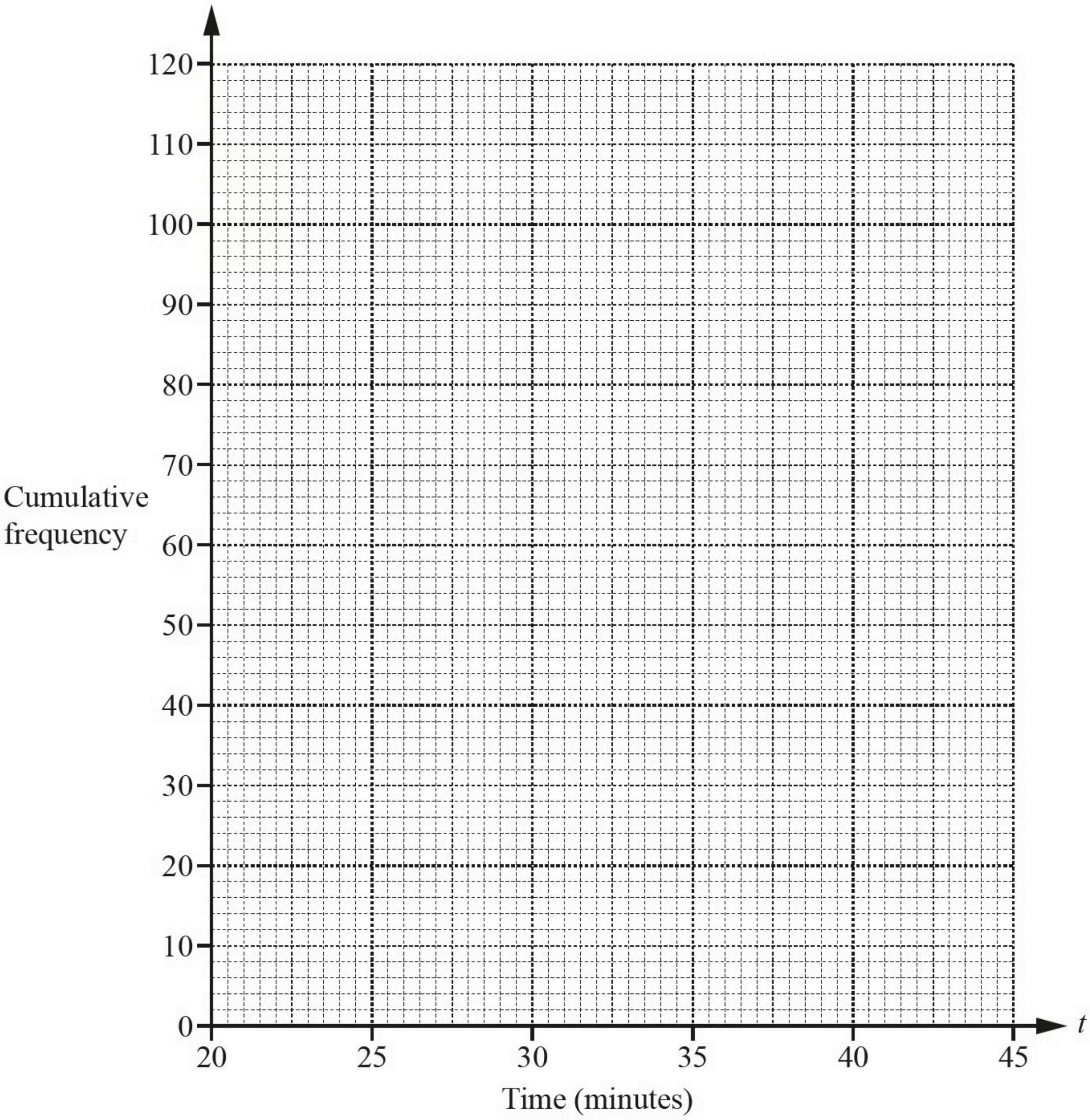
Time (t minutes)	$20 < t \leq 25$	$25 < t \leq 30$	$30 < t \leq 35$	$35 < t \leq 40$	$40 < t \leq 45$
Frequency	44	32	28	12	4

i) Complete the cumulative frequency table.

Time (t minutes)	$t \leq 20$	$t \leq 25$	$t \leq 30$	$t \leq 35$	$t \leq 40$	$t \leq 45$
Cumulative frequency	0	44				

[2]

ii) On the grid, draw a cumulative frequency diagram to show this information.



[3]

iii) Find the median time.

.... min [1]

iv) Find the interquartile range.

.... min [2]

v) Find the number of students who took more than 37 minutes to complete the cooking challenge.

[2]

(10 marks)

6 (a) The table shows information about the speeds of 100 lorries.

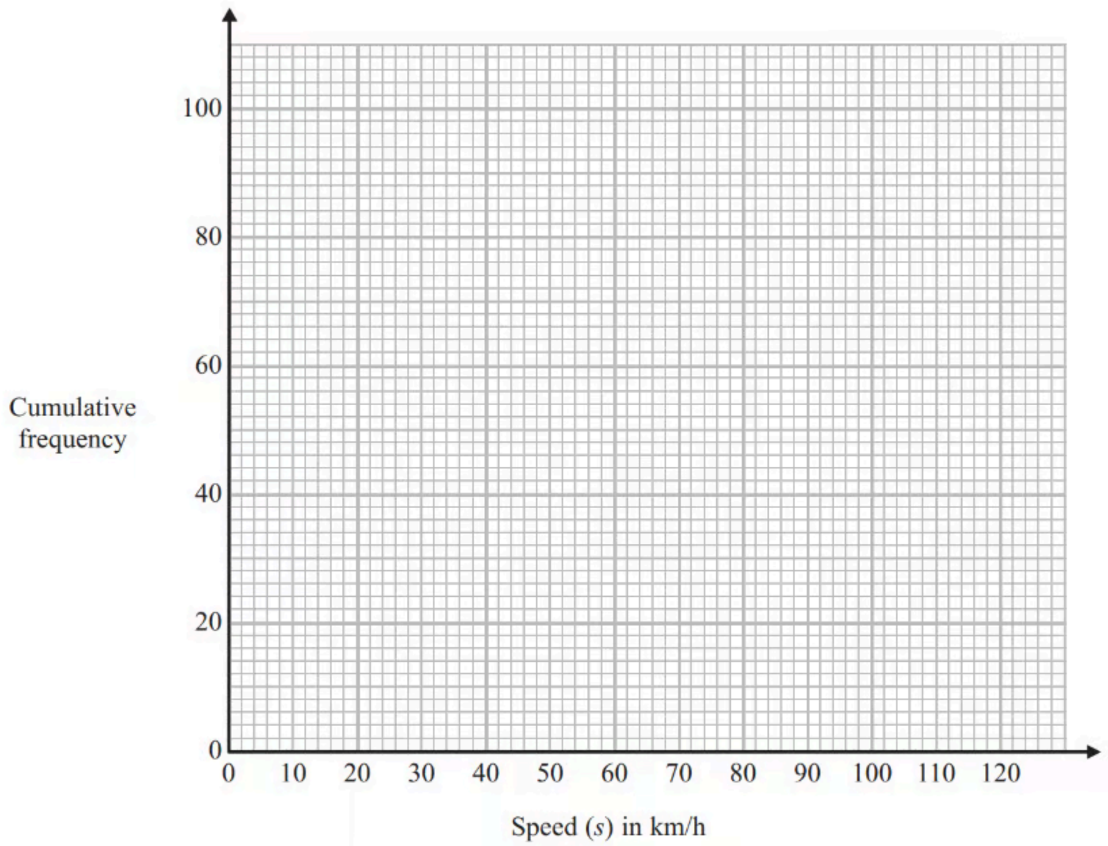
Speed (s) in km/h	**Frequency**
$0 < s \leqslant 20$	2
$20 < s \leqslant 40$	9
$40 < s \leqslant 60$	23
$60 < s \leqslant 80$	31
$80 < s \leqslant 100$	27
$100 < s \leqslant 120$	8

Complete the cumulative frequency table for this information.

Speed (s) in km/h	**Cumulative frequency**
$0 < s \leqslant 20$	2
$0 < s \leqslant 40$	
$0 < s \leqslant 60$	
$0 < s \leqslant 80$	
$0 < s \leqslant 100$	
$0 < s \leqslant 120$	

(1 mark)

(b) On the grid, draw a cumulative frequency graph for your table.



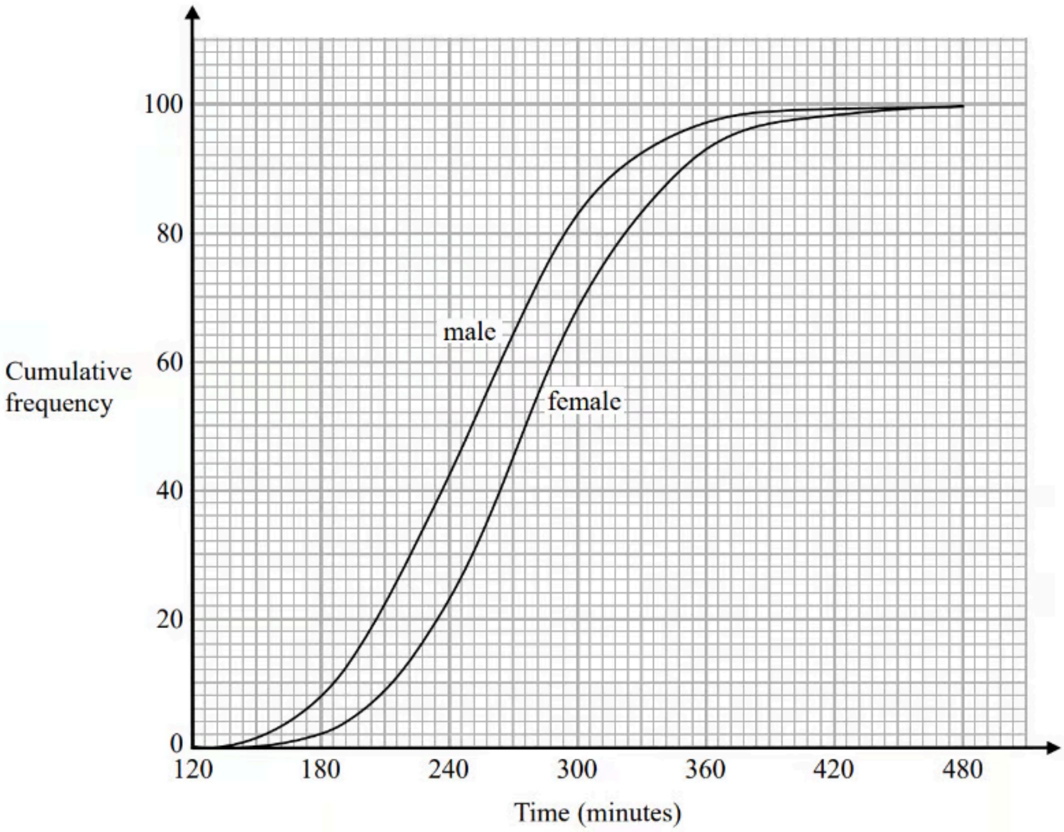
(2 marks)

(c) Find an estimate for the number of lorries with a speed of more than 90 km/h.

(2 marks)

Hard Questions

1 (a) The cumulative frequency graphs show information about the times taken by 100 male runners and by 100 female runners to finish the London marathon.



A male runner is chosen at random.

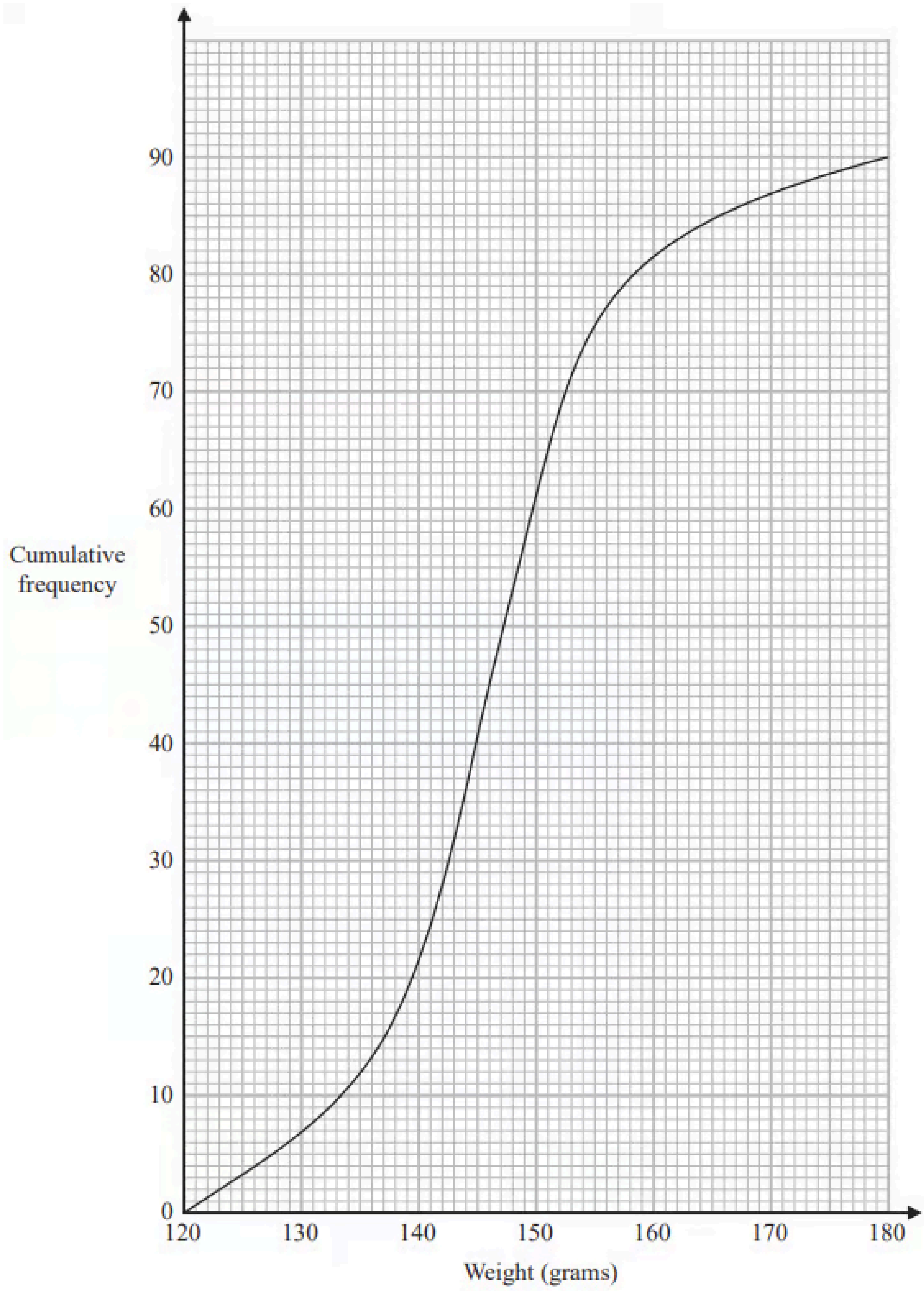
Find an estimate for the probability that this runner took less than 4 hours to finish the London marathon.

(2 marks)

(b) Use medians and interquartile ranges to compare the distribution of the times taken by the male runners with the distribution of the times taken by the female runners.

(4 marks)

2 The cumulative frequency graph gives information about the weights, in grams, of 90 bags of sweets.



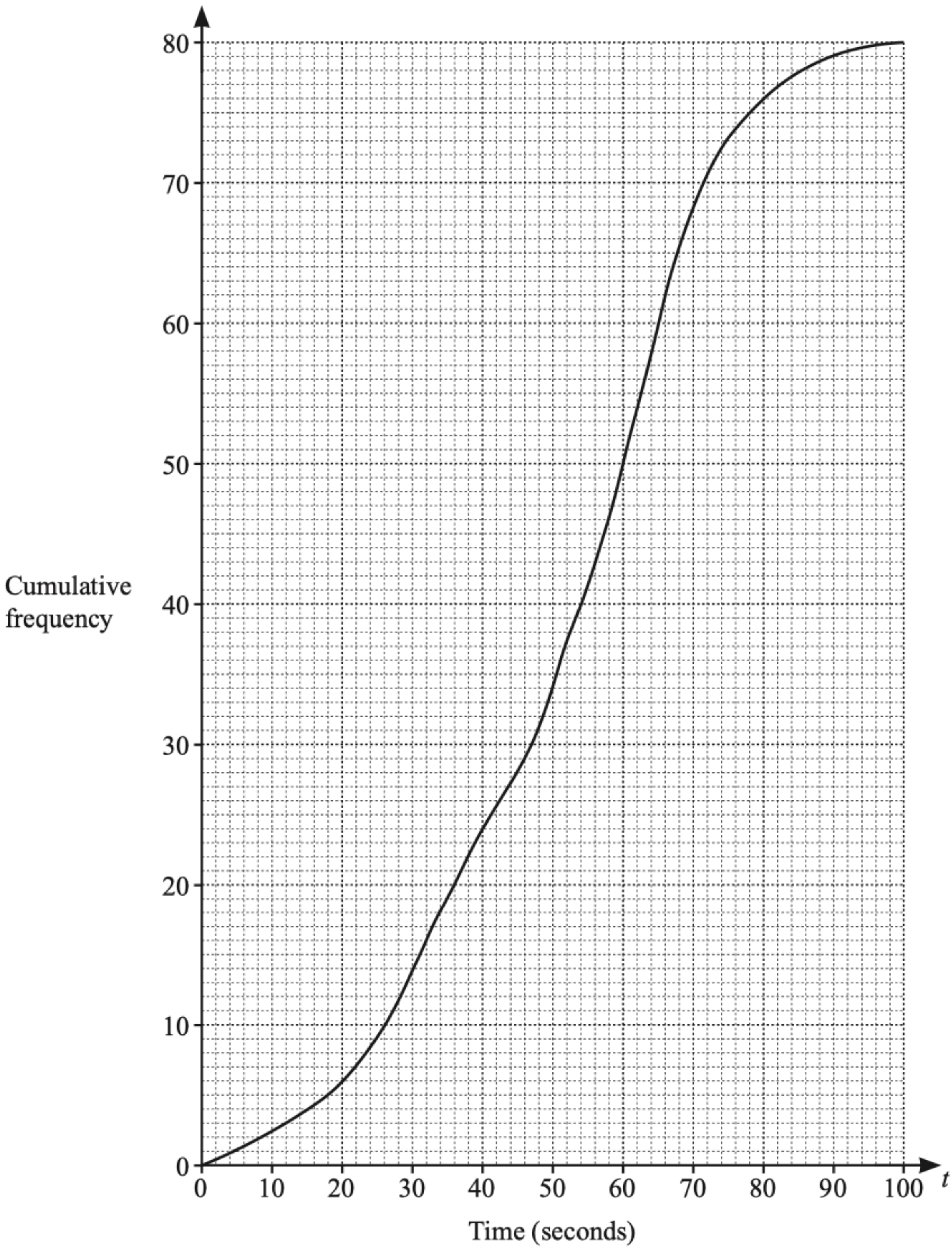
Roberto sells the bags of sweets to raise money for charity.
Bags with a weight greater than d grams are labelled large bags and sold for 3.75 euros each bag.

The total amount of money he receives by selling all the large bags is 93.75 euros.

Find the value of d.

(3 marks)

3 (a) The cumulative frequency diagram shows information about the time taken, t seconds, for a group of girls to each solve a maths problem.



Use the cumulative frequency diagram to find an estimate for

i) the median,

.... s [1]

ii) the interquartile range,

.... s [2]

iii) the 20th percentile,

.... s [1]

iv) the number of girls who took more than 66 seconds to solve the problem.

[2]

(6 marks)

(b) i) Use the cumulative frequency diagram to complete the frequency table.

Time (t seconds)	$0 < t \leq 20$	$20 < t \leq 40$	$40 < t \leq 60$	$60 < t \leq 80$	$80 < t \leq 100$
Frequency	6				4

[2]

ii) Calculate an estimate of the mean time.

.... s [4]

(6 marks)

(c) A group of boys solved the same problem. The boys had a median time of 60 seconds, a lower quartile of 46 seconds and an upper quartile of 66 seconds.

i) Write down the percentage of boys with a time of 66 seconds or less.

.... % [1]

ii) Howard says

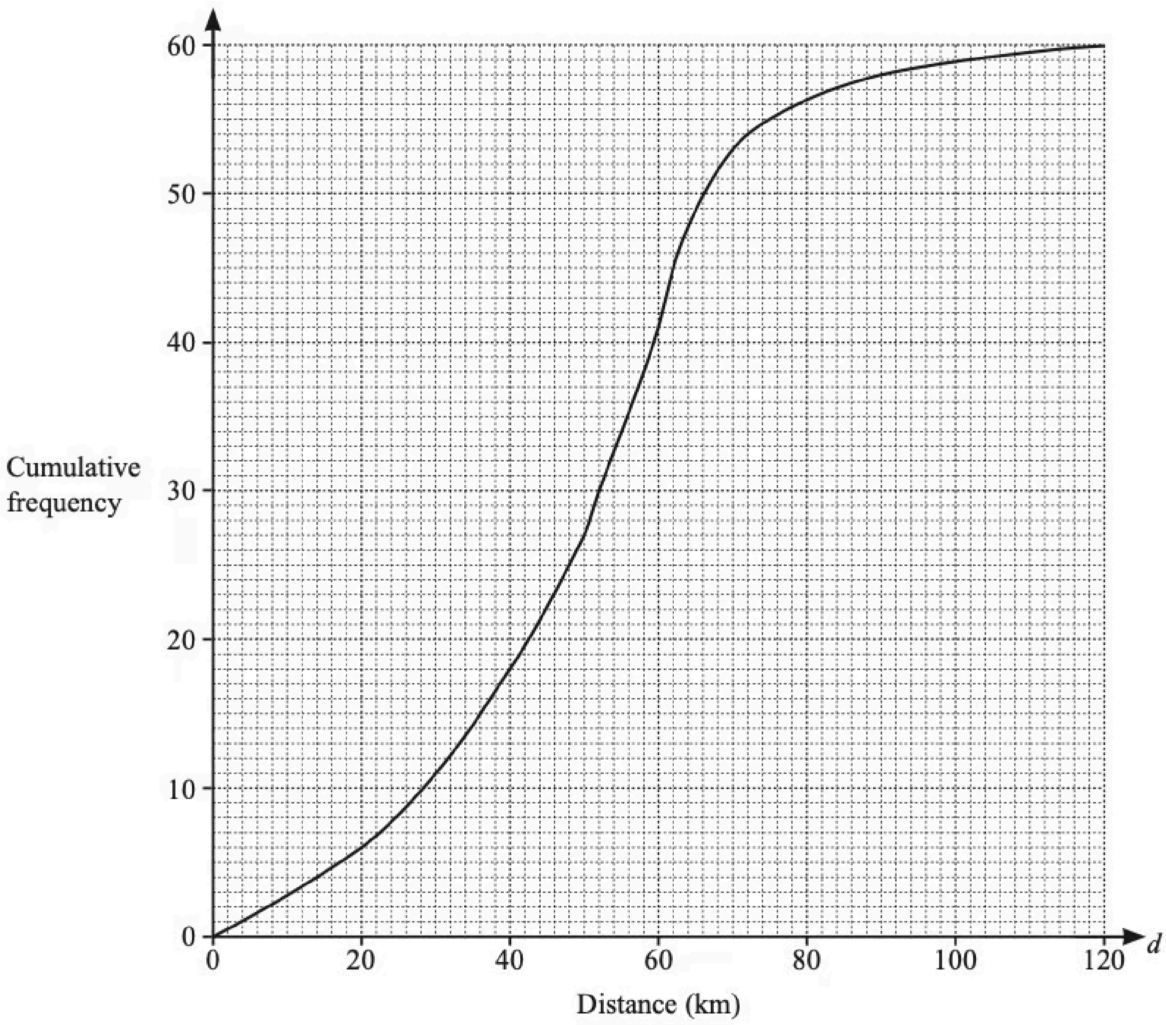
"The boys' times vary more than the girls' times."

Explain why Howard is incorrect.

[2]

(3 marks)

4 (a) The cumulative frequency diagram shows information about the distance, d km, travelled by each of 60 male cyclists in one weekend.



Use the cumulative frequency diagram to find an estimate of

i) the median,

.... km [1]

ii) the lower quartile,

.... km [1]

iii) the interquartile range.

.... km [1]

(3 marks)

(b) For the same weekend, the interquartile range for the distances travelled by a group of female cyclists is 40 km.

Make one comment comparing the distribution of the distances travelled by the males with the distribution of the distances travelled by the females.

(1 mark)

(c) A male cyclist is chosen at random.

Find the probability that he travelled more than 50 km.

(2 marks)

(d) i) Use the cumulative frequency diagram to complete this frequency table.

Distance (d km)	**Number of male cyclists**
$0 < d \leq 40$	18
$40 < d \leq 50$	9
$50 < d \leq 60$	
$60 < d \leq 70$	
$70 < d \leq 90$	
$90 < d \leq 120$	2

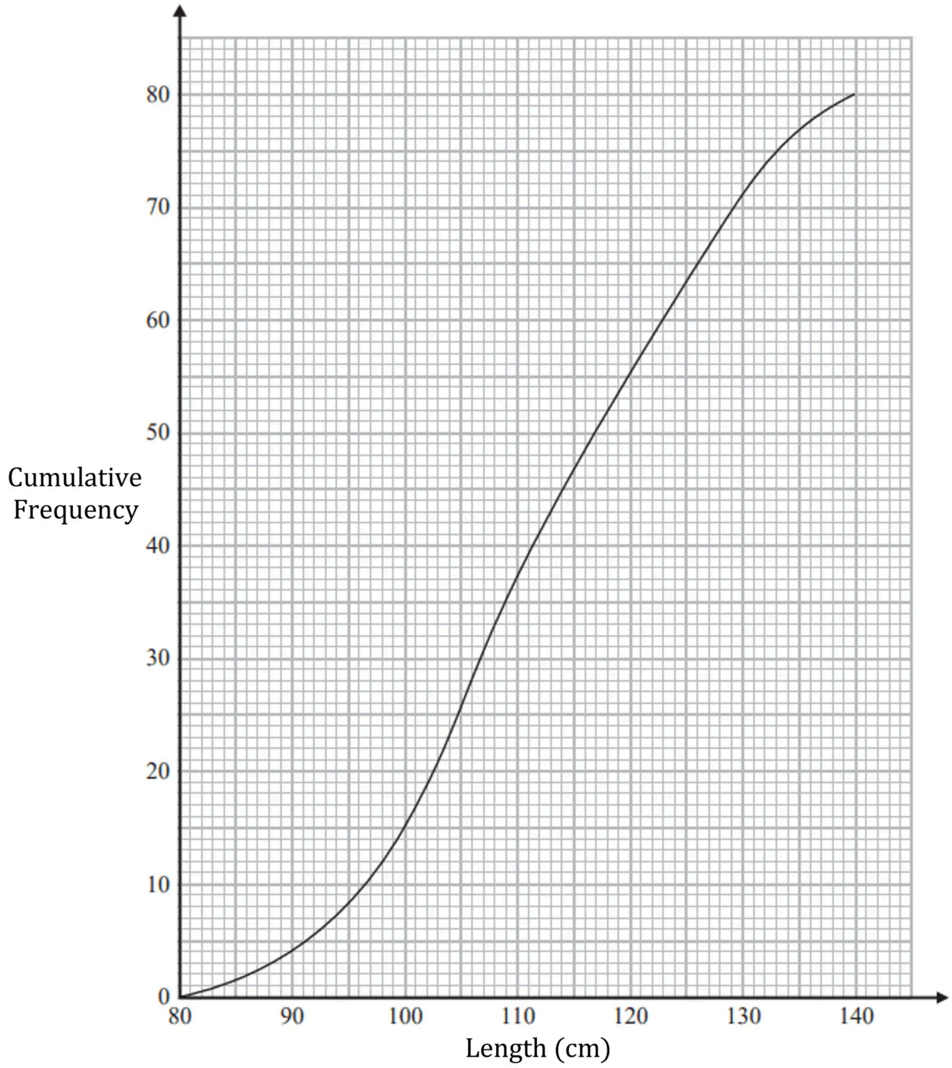
[2]

ii) Calculate an estimate of the mean distance travelled.

.... km [4]

(6 marks)

5 (a) The cumulative frequency graph shows information about the length, in minutes, of each of 80 fish.



Use the graph to find an estimate for the interquartile range of the fish.

(3 marks)

(b) Clint says, “More than 35% of the fish are over 120 cm long.”

Is Clint correct?

Give a reason for your answer.

(4 marks)